

Fractal Information Theory Project

FIN ST357–ST371

Simple-Fold Normal Form, Globality Obstructions,
and Twelve-State Observer Geometry

Local analytical, interval, exact finite, and bounded numerical research

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Repository snapshot. Audited tracked base: `bb540a767e210a5a3bf816e4868e7bcefbac3872`. The working tree contains uncommitted research artifacts. The accompanying SHA-256 ledger fixes the exact release packet.

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Abstract

Programs ST357–ST371 continue the local FIN research program after Release 10.80. The principal positive result upgrades the certified one-null-direction stationary event to a local simple saddle-node theorem. Exact symmetrizability supplies its left nullvector, and outward interval arithmetic separates both transversality coefficients from zero: $w^T F_g \in [-0.359009428, -0.359009343]$ and $w^T F_{xx}[v, v] \in [0.276706627, 0.276707990]$. A separate tangent/transverse Krawczyk argument certifies one continuous local branch slab through the worst-conditioned region of the earlier continuation.

The global-minimizer problem is not closed. The valid root lower bound $V_4 \geq -3.320614558$ is too weak, and a positive rational state is constructed for which averaging with every one of the twelve reflections strictly raises the objective. Thus symmetry, reflection averaging, and the present entropy–quadratic bound do not compose into a global proof. For a supplied rank-seven Fourier mediator, however, a unique local reflection-even minimum is interval-certified; it lies below the uniform state and all pure vertices, while all boundary minimizers are excluded analytically.

Molien averaging shows that the complete D_{12} -invariant mean-zero algebra has dimensions 53 and 365 in degrees four and six, respectively. Hence the previously classified modal energy algebra misses at least 32 quartic and 309 sextic invariant directions. This expands the admissible nonlinear vocabulary but does not source any coefficient.

On the observer side, cyclic twirling reduces the deficiency of arbitrary 12-state convolution experiments to one finite linear program, and an exact rational example proves that the resulting order is genuinely partial. Multinomial recovery bounds scale as $N^{-1/2}$. A strict-heat Markov source conditionally supports numerical block-three and block-four rate–distortion curves. Under an added martingale-difference law, relative multilevel clock drift scales as the square root of depth instead of linearly. No active coupling source or independent laboratory record is present. No selector, absolute units, time arrow, spacetime, physical matter law, legacy-role transfer, Standard Model, gravity, L_{total} , or Theory-of-Everything closure is claimed.

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1 Scope, ontology, and proof discipline

The finite strict operator used below is

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)) \quad (x \neq y),$$

with $A1 = 0$, $A \succeq 0$, and $s = 1.660307278766099$. The zero-diagonal convention is indispensable: the displayed s is the off-diagonal row sum.

The legacy ontological kernel remains an intermediate bridge object. It is not silently substituted for the strict operator, and no historical legacy physical role is transferred. The project ontology—the nadsoliton itself as primordial fractal information in a solitonic state, followed internally by light, matter, and an emergent observer—is preserved as context. It is not a premise in the finite theorems.

[PROVEN] denotes exact analysis or outward interval certification. **[STRONG NUMERICAL EVIDENCE]** denotes reproducible floating computation. **[PARTIAL]** denotes a missing global, continuum, source, or physical step. **[BLOCKED]** marks information that local computation cannot manufacture.

2 Executive result ledger

Pro-gram	Status	Result
ST357	[PROVEN]	The ST343 event satisfies both interval fold transversality conditions and is a local simple saddle-node.
ST358	[PROVEN] ;	A rigorous global/cell lower-bound primitive is derived; it is too loose for
ST359	partial [PROVEN]	global closure.
ST360	[PARTIAL]	One explicit rational state defeats every one-step reflection symmetrization.
ST361	[PROVEN]	The current symmetry–rearrangement–root-bound global strategy is stopped without deciding global uniqueness.
ST362	[PROVEN]	A supplied rank-seven mediator has a unique strict local minimum below uniform and pure states; boundary minima are impossible.
ST363	[PROVEN] ;	Full D_{12} invariant counts are 53 in degree four and 365 in degree six.
ST364	partial [PROVEN]	Normalized Chernoff derivatives are uniformly bounded at all depths; infinite-depth convergence remains open.
ST365	[PROVEN]	A continuous local tangent/transverse branch slab is interval-certified.
ST366	[PROVEN]	For the frozen grid dual there are exactly three simple thresholds and two selected bands.
ST367	[PROVEN]	General cyclic observer deficiency is a 12-variable LP; exact examples prove partial ordering.
ST368	Numerical	Nonasymptotic multinomial upper and two-point lower recovery bounds both have root-count scaling.
ST369	[PROVEN]	Block-three and block-four strict-heat rate–distortion curves are computed; block five is explicitly stopped.
ST370	conditional [BLOCKED]	Martingale log-clock drift has square-root-depth concentration under a supplied stochastic law.
ST371	[BLOCKED]	No new strict typed source for the active coupling is present.
		No independently registered raw-event record is present.

3 ST357: interval-certified simple saddle-node

For reflection-even probabilities $q = (q_0, \dots, q_6)$ with multiplicities $D = \text{diag}(1, 2, 2, 2, 2, 2, 1, 1)$, write the stationarity system as

$$F_i(q, \lambda, g) = \log(12q_i) + 1 - g(Bq)_i + \lambda = 0, \quad \sum_i D_{ii}q_i = 1.$$

ST343 certified a unique root of the augmented system

$$F(x, g) = 0, \quad F_x(x, g)v = 0, \quad \frac{1}{2}(\langle v, v \rangle - 1) = 0$$

inside a 17-dimensional box, with exactly a one-dimensional kernel of F_x . The weighted Jacobian DF_x is symmetric. Therefore, if v spans the right kernel, $w = Dv$ is an exact left nullvector.

Outward evaluation on the entire ST343 root box gives

$$w^T F_g \in [-0.35900942708142014, -0.3590093430691903],$$

$$w^T F_{xx}[v, v] \in [0.2767066271841476, 0.27670798909386124].$$

Their product lies in $[-0.099340776633439, -0.0993402644482722]$.

Theorem 3.1 (local simple-fold normal form). *At the unique ST343 augmented root, the stationary solution set is locally smoothly equivalent to a simple saddle-node: in suitable coordinates its scalar reduced equation has nonzero parameter and quadratic coefficients.*

This is a local theorem about the supplied negative-information functional. It does not prove global branch topology, dynamic stability, a physical phase transition, or a strict source for g .

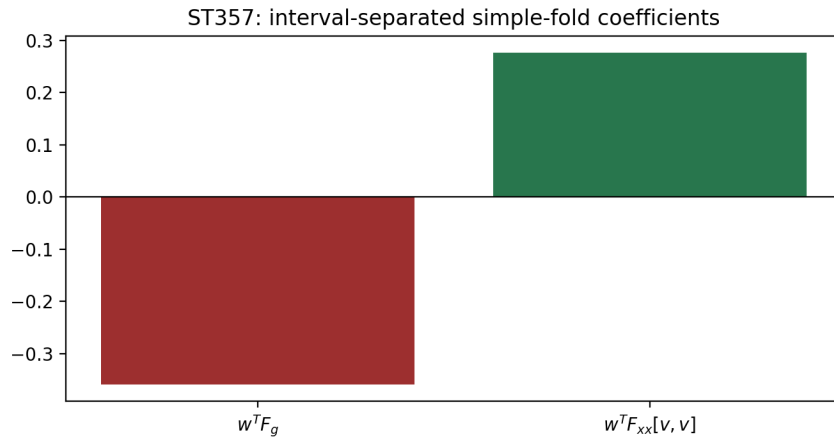


Figure 1: The two fold coefficients are separated from zero by outward intervals.

4 ST358–ST360: why the global proof still fails

Let $u = (1/12, \dots, 1/12)$ and

$$V_g(p) = \sum_i p_i \log(12p_i) - \frac{g}{2} \langle p - u, A(p - u) \rangle.$$

Relative entropy is nonnegative. Since $A \succeq 0$, the quadratic is convex, so its maximum over the probability simplex occurs at a vertex. Circulant symmetry makes the vertex value equal to s . Hence

$$V_g(p) \geq -\frac{gs}{2}.$$

At $g = 4$ this gives $V_4 \geq -3.320614557532198$, while the localized candidate is near -0.84519850768 . The gap 2.4754 is too large for a root-box certificate.

There is nevertheless a rigorous branch-and-bound primitive. On a box $p_i \in [\ell_i, u_i]$, minimize each separable term $p_i \log(12p_i)$ at the clipped point $1/(12e)$, and maximize the convex quadratic at vertices of the box–simplex polytope. This supplies a valid cell lower bound, not a practical global closure yet.

ST359 then tests the tempting reflection rearrangement. Let p have integer numerators

$$(500, 1, 3, 2, 1, 470, 1, 1, 1, 11, 1, 1)$$

and denominator 993. For every dihedral reflection R_k , outward strict-operator evaluation proves

$$V_4\left(\frac{p + R_k p}{2}\right) - V_4(p) > 0.$$

The smallest certified increase is 0.08893328445712058.

Theorem 4.1 (one-step reflection rearrangement no-go). *There is no general theorem asserting that at least one one-step reflection average is nonincreasing for the declared V_4 .*

Together with ST344, this stops the low-cost strategy “symmetry + reflection averaging + current root bound.” It neither proves nor refutes global uniqueness modulo D_{12} . A future proof needs a sharper coupled entropy–quadratic envelope or an exact global dual.

5 ST361: a certified rank-seven local minimum

Retain the complete Fourier modes $k = 3, 4, 5, 6, 7, 8, 9$ of the strict circulant operator. This gives the first covariant rank at which the pure-state energy beats the uniform state at $g = 4$. A radius- 10^{-8} Krawczyk box contains exactly one positive reflection-even stationary point. Its inclusion margin is 9.9963×10^{-9} , and the rigorous tangent Hessian lower bound is 5.78807.

At the exact root, the objective enclosure is strictly below both zero (uniform state) and the pure-vertex enclosure. Moreover, no boundary point can be a minimizer of V_g : adding ε to a zero coordinate, with compensating removal from a positive coordinate, produces an $\varepsilon \log \varepsilon$ entropy term whose negative directional divergence dominates the finite derivative of the quadratic.

Theorem 5.1 (rank-seven localized local minimum). *For the supplied rank-seven mediator and $g = 4$, the certified root is a strict interior local minimum; it beats the uniform state and every pure vertex, and no boundary point is a global minimizer.*

The remaining competitors are interior and may have trivial stabilizer. Therefore this is not a global-minimum theorem and does not source a physical mediator.

6 ST362: the full nonlinear invariant vocabulary

For the permutation representation of D_{12} on twelve coordinates, removing the trivial line multiplies the Molien series by $(1 - t)$. Thus

$$M(t) = \frac{1-t}{24} \sum_{g \in D_{12}} \prod_{C \in \text{cycles}(g)} (1 - t^{|C|})^{-1}.$$

Exact series expansion gives, in degrees zero through eight,

$$(1, 0, 6, 12, 53, 124, 365, 807, 1892).$$

Theorem 6.1 (strictness of the modal-energy subalgebra). *The complete mean-zero D_{12} -invariant spaces have dimension 53 at degree four and 365 at degree six. Since the modal-energy subalgebra has dimensions 21 and 56, it misses at least 32 quartic and 309 sextic invariant directions.*

This makes phase-sensitive or nonspectral nonlinearities mathematically available. It does not choose an invariant, determine its coefficient, break D_{12} , or discharge the selector obstruction. Constructing a sparse Reynolds basis is deferred.

7 ST363–ST365: continuation and infrared selection

For a binary hidden-state transfer tree, let $T_n(s)$ be the depth- n Chernoff transfer. Differentiation under the finite sum yields

$$\frac{1}{n} \partial_s \log T_n(s) = \mathbb{E}_s \left[\frac{1}{n} \log \frac{P_0(Y_{1:n})}{P_1(Y_{1:n})} \right].$$

In the declared emission fixture, every one-step likelihood ratio lies in $[1/49, 49]$. Consequently

$$-\log 49 \leq \frac{1}{n} \partial_s \log T_n(s) \leq \log 49$$

for all depths and initial beliefs. This prevents linear growth of the normalized derivative but does not prove convergence to an infinite-depth pressure derivative. The depth-4 through depth-12 collocation is diagnostic only.

ST364 changes coordinates at the worst-conditioned ST298 sample: one coordinate follows the tangent, and seven span its orthogonal complement. A Taylor-form interval Krawczyk map certifies that for every tangent parameter in $[-2 \times 10^{-7}, 2 \times 10^{-7}]$ there is one unique transverse correction in the radius- 10^{-7} ball. The minimum inclusion margin is 3.9530×10^{-8} and the maximum transverse contraction row sum is 0.477833. This is a genuine continuous local slab, not yet an overlapping cover of the complete branch.

For the frozen 1000-grid ST350 dual, use $y = 2\mu$ and write its analytic threshold as

$$G(y) = \frac{2a_1 y}{1 + e^{-y}} + \frac{8a_4 y e^{-3y}}{1 + e^{-4y}} - \nu.$$

An exhaustive outward sign cover, interval derivative checks, and a positive analytic tail prove exactly three simple positive roots. Their μ boxes are

$$\begin{aligned} &[0.0142283044721, 0.0142283046722], \\ &[1.0497646833616, 1.0497646835616], \\ &[1.2738113230865, 1.2738113232865]. \end{aligned}$$

Therefore the frozen selector has exactly two selected bands. This corrects the ST350 language: four grid entry/exit transitions had been mislabeled as four bands. The dual multipliers remain finite-grid numerical optimizers; continuum optimality is not certified.

8 ST366–ST367: observer comparison and recovery

Let C_p be a 12-state cyclic convolution experiment with noise law p . Averaging any postprocessing kernel over cyclic shifts cannot increase its worst-row total variation and produces another convolution. Hence

$$\delta(C_p, C_q) = \min_{r \in \Delta_{11}} \text{TV}(p * r, q),$$

a linear program in twelve filter weights plus absolute-value slacks.

Theorem 8.1 (cyclic deficiency reduction). *The displayed finite LP gives the exact one-way Le Cam deficiency for any pair of declared 12-state cyclic convolution experiments.*

For one constructed garbling, the forward deficiency is zero and the reverse numerical LP value is 0.1792857. A second rational pair is incomparable: exact inversion of one circulant through the other yields minimum filter coefficients $-505/4392$ and $-1437467/4404309$ in the two directions. Thus no stochastic convolution filter exists either way. The positive displayed deficiency magnitudes, 0.2 and 0.3, are numerical LP values; incomparability itself is exact.

For N independent correctly labelled observations of a 12-category fiber law, coordinate Hoeffding and a 24-event union bound give, with confidence $1 - \alpha$,

$$\|\hat{p} - p\|_2 \leq \sqrt{\frac{12 \log(24/\alpha)}{2N}}.$$

A two-point construction with categorical KL divergence, product Pinsker, and Le Cam’s lemma gives a matching order- $N^{-1/2}$ expected-risk lower bound. These are operational sample-complexity theorems conditional on ideal iid labels, not laboratory evidence.

9 ST368: conditional strict-heat rate–distortion

Supplying $\tau = 0.5$ and the Markov interpretation gives $P = \exp(-\tau A)$. For normalized block Hamming distortion, Kronecker-factorized Blahut–Arimoto avoids constructing the 12^{2n} distortion matrix. Complete numerical curves were computed for blocks three and four. At fixed block length, the zero-distortion limit is $H(X_{1:n})/n$, equal here to 2.932144448 bits/symbol for $n = 3$ and 2.850542191 bits/symbol for $n = 4$. These block entropies converge to the exact conditional Markov entropy rate 2.605735421 bits/symbol only as $n \rightarrow \infty$.

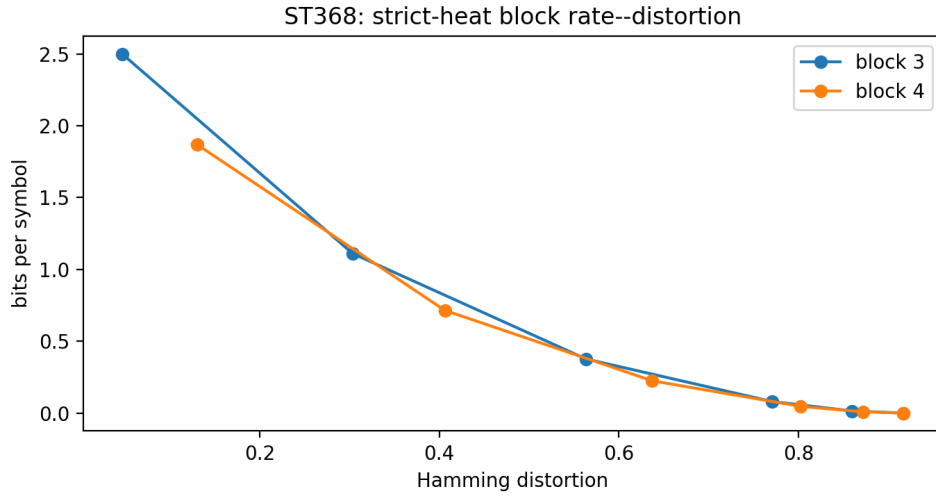


Figure 2: Numerical block-three and block-four rate–distortion curves for the supplied strict-heat source.

Block five is explicitly stopped: source storage already contains $12^5 = 248,832$ entries, and it is unnecessary for this low-compute batch. The finite curves do not prove the asymptotic rate–distortion function. More importantly, τ , the Markov reading, and the distortion remain supplied operational choices rather than strict-derived physics.

10 ST369: relative clocks under stochastic refinement error

ST355 proved adversarial product bounds. ST369 adds a distinct conditional question. If mode-pair log-error differences X_n form a martingale-difference sequence with $|X_n| \leq b_n$, then Azuma–Hoeffding gives

$$\Pr\left(\left|\sum_{n=1}^L X_n\right| \geq \sqrt{2 \log(2/\alpha) \sum_{n=1}^L b_n^2}\right) \leq \alpha.$$

Thus fixed-amplitude stochastic error grows as $\sqrt{L}$ at fixed confidence, whereas the adversarial bound grows as L . A common calibration gauge still cancels in relative clock ratios. For the wave channel, the log-frequency error is half the corresponding generator eigenvalue log-error.

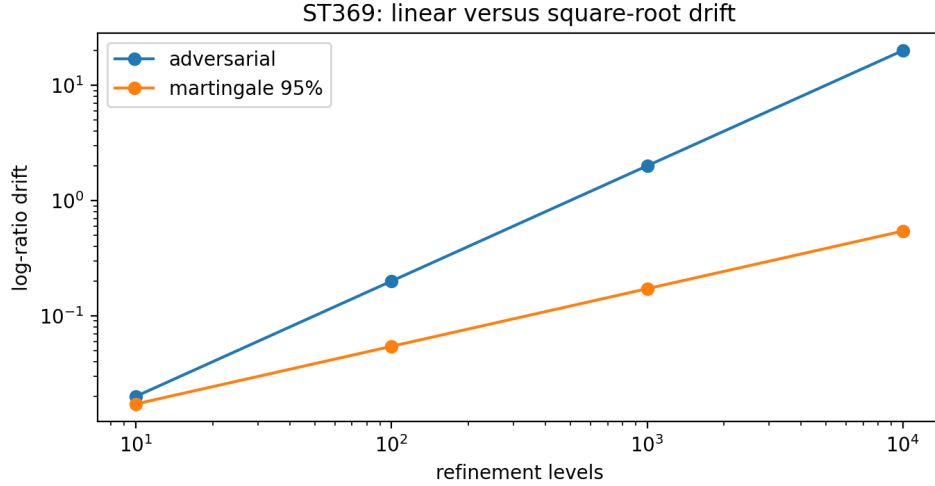


Figure 3: Conditional martingale concentration versus the adversarial accumulation bound.

The centering and boundedness law is an additional stochastic premise. This result creates neither seconds nor a direction of time. Under unitary phase evolution, $(A, t) \mapsto (cA, t/c)$ preserves phases; under wave evolution $\cos(t\sqrt{A})$, the corresponding invariance is $(A, t) \mapsto (cA, t/\sqrt{c})$. One fixed physical dimension assignment cannot make A simultaneously a $1/t$ generator and a $1/t^2$ stiffness without channel-specific maps and calibration.

11 ST370–ST371: two honest stops

ST370 searched the declared local admission channel for a new strict typed active-coupling source satisfying strict provenance, nonzero signed coupling, D_{12} typing, selector safety, kernel-split safety, and absence of a hidden dimensional premise. No candidate is present. Closed source lanes were not replayed. This is a bounded artifact-state result, not proof that no enlarged theory can contain such a source.

ST371 likewise finds no independently registered raw-event bundle. Local code cannot generate independent custody, apparatus calibration, blinded registration, unblinding, or empirical events. This is an evidentiary stop, not a failed experiment.

12 Synthesis: what changed, and what did not

Three advances survive falsification in this batch:

- The negative-information stationary event is now locally classified as a simple saddle-node with interval-certified coefficients.
- The current global proof strategy has a constructive, not merely rhetorical, boundary: generic symmetry reduction is nonexhaustive, every reflection average can worsen the objective, and the available global lower bound is quantitatively insufficient.
- Observer comparison extends from one symmetric confusion family to arbitrary cyclic convolution experiments, where exact finite deficiency geometry is available and need not be a total order.

The deeper interpretation remains restrained. The same finite operator can support several mathematical channels, and relational quantities can survive representation or common calibration changes. That is compatible with information being encoded across different media, but it does not establish medium-free physical information or prove that the universe is a projection of FIN. Spectrum alone remains insufficient: projectors, preparations, instruments, records, and composition laws are operationally relevant.

The strong symmetry of the strict core provides a reservoir of equivalent possibilities, but symmetry is not itself positive feedback. Selection requires a state-dependent instability, orbit choice, boundary condition, measure, or other symmetry-breaking object. No such strict-sourced selector is exported here, and QW-2191 remains open.

13 Recommended next studies: ST372–ST386

Pro-gram	Objective	Acceptance criterion	Priority
ST372	Overlapping rotated slabs	Certify a connected interval cover across every failed ST349 section, including overlap maps.	High
ST373	Sharper global cell envelope	Derive a joint entropy–quadratic convex envelope whose root-cell bound closes at least 80% of the current gap.	Highest
ST374	Exact global dual	Construct an interval-admissible dual certificate below the localized value, or return a bounded no-go.	Highest
ST375	Global orbit theorem	Only after ST373/374, prove or refute uniqueness modulo D_{12} without assuming fixed symmetry.	High
ST376	Rank-seven interior cover	Exclude all other interior rank-seven critical basins using interval boxes and Morse data.	High
ST377	Quartic Reynolds basis	Export an explicit independent basis of all 53 quartic invariants and separate spectral from phase-sensitive terms.	High
ST378	Sparse sextic module	Construct a computationally usable subset/basis certificate for the 365 sextic invariants.	Medium
ST379	Chernoff spectral gap	Prove a Lasota–Yorke/cone contraction estimate for the normalized transfer derivative, or exhibit a counterexample.	Highest
ST380	Continuum IR dual KKT	Replace the frozen grid dual with interval-certified continuum multipliers and roots.	Highest
ST381	Fourier bounds for deficiency	Derive analytic lower bounds for cyclic convolution deficiency and characterize zero-deficiency exactly.	High
ST382	Recovery with nuisance	Add calibrated loss, label error, and nuisance parameters to the multinomial minimax theorem.	Medium
ST383	Asymptotic rate–distortion	Use transfer operators or certified block bounds; do not infer an asymptote from block five alone.	Medium
ST384	Dependent clock noise	Replace the martingale premise by mixing/dependence bounds and test scale-uniformity.	Medium
ST385	Typed-source admission	Run only if a genuinely new strict-sourced active or selector object is supplied; otherwise preserve the stop.	Condi-tional
ST386	Independent evidence gate	Run only on a preregistered external raw-event packet with custody and calibration metadata.	External

The recommended local order is $ST373 \rightarrow ST374 \rightarrow ST379 \rightarrow ST380$. ST372 and ST377 are valuable parallel mathematical tracks. ST385 and ST386 must not be simulated from existing artifacts.

14 Reproducibility and provenance

The executable packet contains:

- `fin_st357_st371_research.py`, which regenerates all fifteen program packets, the aggregate JSON, CSV summary, and three figures;
- `test_fin_st357_st371.py`, containing sixteen live/serialized tests;
- one JSON packet per program, the aggregate `FIN_ST357_ST371_Results.json`, and a SHA-256 ledger;
- this TeX source and its compiled PDF.

The replay used Python 3, NumPy, SciPy, SymPy, mpmath interval arithmetic, and Matplotlib. The random seed is 20260818, although the principal interval and exact finite results do not depend on randomness. All sixteen tests pass.

Internal dependencies are explicit: ST357 consumes the certified ST343 box; ST358/ST359 use the strict interval family; ST361 uses the earlier rank threshold and localization routines; ST363 continues the declared synthetic hidden-state fixture; ST364 consumes the ST298 path; ST365 freezes the corrected ST350 1000-grid dual; ST368 consumes the strict heat operator; ST369 continues the ST355 drift decomposition. These are repository results, not independent external validations.

15 Final verdict

Release 10.81 closes one important local mathematical question: the certified stationary event is a simple saddle-node. It also makes the central remaining variational obstruction more precise. The difficulty is no longer locating a plausible localized state; it is proving globality without illicit symmetry reduction and without a weak separable bound.

The observer results strengthen the operational mathematics of finite information transfer, but they also reinforce the physical boundary. An operator, even with dual dynamics and multiscale relational clocks, does not determine preparations, instruments, environment, apparatus, record, absolute scale, or empirical meaning. Those objects must be derived from a new typed principle or supplied as additional axioms and tested independently.

Accordingly, the scientifically defensible present description is: a coherent finite spectral–variational–informational research framework with several certified local and operational theorems, not a completed physical theory and not a Theory of Everything.

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